ALADIN audit, Phase 3: what works in Indian markets (evidence review)

Written 2026-11-08. How I searched: OpenAlex (the open scholarly database, official API) for 23 searches across options A–G, about 280 records (`aladin-audit/phase3/search.json` , made by `scripts/phase3_search.py`); Semantic Scholar for missing abstracts; the IIM Ahmedabad data library page and its data files. Web search engines answered with a bot check, so I did not use them.

What "read" means here. For most papers I read the **abstract** (the authors' own summary), not the full text; the table says which. I list a source only if I opened at least its abstract. Two papers had no abstract anywhere and are listed as "not read", not used. NSE's factor-index factsheets (niftyindices.com) answered with a script-only page, so they are **not read**.

Grades. Strong = many stocks and years, India, and either held on later data or includes costs. Medium = India and a reasonable sample, but no costs or no later test. Weak = small sample, short period, no trading test, or not India.

3.1 and 3.2 Sources and grades

#	Source (link)	Read	India sample	Costs	Held later?	Finding	Grade
1	IIM Ahmedabad Fama-French and Momentum data library, release to **December 2025** (not March 2023 as the brief said) — https://faculty.iima.ac.in/iffm/Indian-Fama-French-Momentum/	page + monthly data files	BSE+NSE, micro-caps and penny stocks removed, survivorship-adjusted, Oct 1993–Dec 2025	no	**yes**: the method was published in 2013 and momentum kept working 2014–2025	Winners-minus-losers +10.6% a year since 1994 (+11.5% 2016–2025). Long-only big-cap winners +18.6% a year 2016–2025 vs market +9.2%; worst fall −26.5% vs −34.4%. Crash months −27.6% (Nov 2001), −25.0% (May 2009)	**Strong**
2	Implementing a systematic long-only momentum strategy: evidence from India (SSRN 2019) — https://doi.org/10.2139/ssrn.3510433	abstract	NIFTY 100, top tenth, monthly	**yes** (discount broker)	–	+10.7% a year over the NIFTY 100; 32% turnover a month; occasional crashes	Medium
3	Momentum anomaly: evidence from the Indian stock market (J. Adv. Mgmt Research 2017) — https://doi.org/10.1108/jamr-11-2015-0081	abstract	470 BSE stocks, 1997–2013	no	no	Momentum profitable; **driven by winners**, so long-only works; not a size or liquidity effect	Medium

#	Source (link)	Read	India sample	Costs	Held later?	Finding	Grade
4	Are prominent equity market anomalies in India fading away? (Global Business Review 2019) — https://doi.org/10.1177/0972150918811248	abstract	NSE 500, 2005–2016	no	–	**Disagrees**: value and momentum explained by risk models in this period; anomalies fading	Medium
5	Risk-managed momentum: evidence from Indian stock market (FIIB Business Review 2021) — https://doi.org/10.1177/23197145211023001	abstract	450 BSE stocks	no	–	Momentum has rare severe crashes; scaling it by risk **doubles** the risk-adjusted return	Medium
6	Equity market anomalies, VIX and asset pricing: trading strategies for India (Indian Economic Journal 2019) — https://doi.org/10.1177/0019466220951978	abstract	2001–2019	no	–	Size, value, profitability, investment, momentum all significant; only size reverses with VIX	Medium
7	The cross-section of Indian stock returns: evidence using machine learning (Applied Economics 2021) — https://doi.org/10.1080/00036846.2021.1982132	abstract	**all listed firms, survivorship-free**, 1994–2019	not stated	**rolling out-of-sample**	14 of 35 characteristics significant; machine learning beats linear forecasts; strategies on the forecasts earn significant returns	**Strong** (no costs stated)
8	Quality investing in the Indian stock market (Managerial Finance 2018) — https://doi.org/10.1108/mf-07-2017-0248	abstract	BSE 500	no	–	Two of four quality measures (Grantham quality, gross profitability) beat the market after size, value, momentum	Medium
9	Machine-learning quality minus junk (Finance Research Open 2025) — https://doi.org/10.1016/j.finr.2025.100041	abstract	BSE non-financials, 2013–2022	no	–	Quality premium exists but **weakens after COVID**	Medium
10	Does profitability explain the low-risk anomaly in India? (Quarterly Rev. Econ. Finance 2025) — https://doi.org/10.1016/j.qref.2025.102060	abstract	NIFTY 500, 2003–2022	no	–	Low-risk stocks do not earn less; profitable stocks do better	Medium
11	The volatility effect: evidence from India (Applied Finance Letters 2016) — https://doi.org/10.24135/afl.v5i1.32	abstract	2001–2015	no	–	Low-volatility tenth +11.4% a year excess vs high +1.3%	Medium
12	Volatility effect and the role of firm quality (IIMB Management Review 2017) — https://doi.org/10.1016/j.iimb.2017.01.001	abstract	BSE 500, 2000–2013	no	–	**Disagrees**: no low-volatility anomaly; high-volatility stocks did better	Medium

#	Source (link)	Read	India sample	Costs	Held later?	Finding	Grade
13	A first look at the properties of India's volatility index (Int. J. Emerging Markets 2012) — https://doi.org/10.1108/17468801211209938	abstract	India VIX	–	–	India VIX carries information about future market volatility	Medium
14	Post-earnings-announcement drift anomaly in India (Theoretical Economics Letters 2018) — https://doi.org/10.4236/tel.2018.814197	abstract	2002–2017	no	sub-periods	Drift significant and robust	Weak (low-tier journal)
15	PEAD and value-glamour anomalies in NSE firms (Trends Econ. Mgmt 2021) — https://doi.org/10.13164/trends.2021.37.27	abstract	100 NSE firms, 2014–2018	no	–	**Disagrees**: stocks properly priced after results; no inefficiency	Weak (small)
16	Price, earnings and revenue momentum: Indian evidence (2015) — RePEc, no DOI	abstract	BSE 500, 2002–2010	no	–	Earnings momentum the most profitable long-short; it subsumes price momentum	Medium
17	Impact of corporate earnings announcements (European Economic Letters 2026) — https://doi.org/10.52783/eel.v16i1.4174	abstract	BSE 500, 2015–2024	no	–	Post-announcement drift present; bad news reacts more strongly	Weak
18	Optimizing LSTM for time series prediction in the Indian stock market (Procedia CS 2020) — https://doi.org/10.1016/j.procs.2020.03.257	abstract	prices	no	no	Price forecasting accuracy only, no trading test	Weak
19	Ensemble machine learning for Indian banks (J. Risk Fin. Mgmt 2022) — https://doi.org/10.3390/jrfm15080350	abstract	bank stocks	no	no	Error measures only, no trading test	Weak
20	Pair trading with cointegration for the Indian market (ASIANCON 2022) — https://doi.org/10.1109/asiancon55314.2022.9909455 and a 2023 preprint https://doi.org/10.36227/techrxiv.23539989	abstract	a few sectors, 2018–2022	no	1 year	Pairs can be found; needs short selling	Weak
21	Investor sentiment and asset returns: the case of India (Afro-Asian J. Fin. Acc. 2017) — https://doi.org/10.1504/aajfa.2018.089198	abstract	2004–2016	–	–	**Sentiment index fails to predict market returns**; only small low-priced stocks	Medium
22	Aggregate news sentiment and stock market returns in India (J. Risk Fin. Mgmt 2023) — https://doi.org/10.3390/jrfm16080376	abstract	market level	–	–	News sentiment effect is **short-lived**	Medium
23	Twitter sentiment and the Indian stock market (2025) — https://doi.org/10.52783/jisem.v10i3.5768	abstract	2015–2022	–	–	No causal link from Twitter sentiment to returns	Weak

#	Source (link)	Read	India sample	Costs	Held later?	Finding	Grade
24	Physical momentum in the Indian stock market (arXiv 2023) — https://doi.org/10.48550/arxiv.2302.13245	abstract	NSE 500, 2014–2021	no	–	Strong **daily reversal** effect	Weak
–	Risk-adjusted returns from statistical arbitrage in Indian stock futures (Asia-Pacific Fin. Markets 2020) — https://doi.org/10.1007/s10690-020-09317-1	**not read** (no abstract)					not used
–	Momentum, reversals and liquidity: Indian evidence (Pacific-Basin Finance J. 2023) — https://doi.org/10.1016/j.pacfin.2023.102193	**not read** (no abstract)					not used
–	NSE factor indices (NIFTY200 Momentum 30, Quality 30, Low Volatility 30) — niftyindices.com	**not read** (script-only page)					not used; can be saved by hand from a browser if you want them

ALADIN's own evidence (from Phase 2, 2015–2026, after 37 bps costs, vs the market, survivorship not removed): the weekly SELL list (bottom 2%) lagged the market by −89 bps a week on 2021–2026; ALADIN 1's best 5% at a 20-day hold earned +56 bps per trade (+42 on 2021–2026); 5- and 10-day holds did not survive costs once the filter was loosened.

3.3 Summary per option

A. Momentum + quality ranking — evidence: **strong for momentum, medium for quality**

Momentum is the best-documented effect in India: the IIMA data show it kept working for 12 years after it was published, it is driven by **winners** (so it works long-only), and a long-only version of the 100 most liquid stocks beat its index by about 11% a year with costs considered. Against it: one study finds it was explained by risk in 2005–2016, and it has rare crashes (−25% to −28% in a month, after sharp market rebounds). Quality helps in some studies but weakened after COVID. Evidence runs to December 2025. Fit with your −25% limit: the long-only winners' worst fall in 2016–2025 was −26.5%, so it needs risk control (E).

B. Earnings-surprise drift (PEAD) — evidence: **mixed, weak**

Two weak papers find drift after results; one small study finds none; a medium study finds earnings momentum the strongest long-short signal (2002–2010). India-specific evidence is mostly from low-tier journals and short samples. It also needs point-in-time result dates: ALADIN does not store filing dates today (it does keep a board-meeting file for the movers test). Confidence: low.

C. Machine-learning probability model — evidence: **one strong study, many weak**

The one high-quality study (survivorship-free, all listed firms, 1994–2019, rolling out-of-sample) finds machine learning beats simple models and its forecasts earn significant returns, but costs are not stated in the abstract. Most other Indian ML papers only measure price-forecast error, which says nothing about profit. ALADIN's own LightGBM ranking is real but small after costs, better at a 20-day hold. Confidence: medium; the overfitting risk is high.

D. Regime-gated blend — evidence: **weak**

I found no Indian study that tests switching factor weights by market state with a trading result. Studies show India has calm and turbulent states (Markov-switching papers) and that most anomalies do not reverse with VIX (source 6). Confidence: low.

E. Risk-first model (predict risk, then size and filter a simple ranking) — evidence: **medium**

India VIX carries information about future volatility (source 13), and scaling momentum by risk doubled its risk-adjusted return in India (source 5). Risk is more predictable than direction, and this is the option that most directly serves your −25% limit. Confidence: medium as an overlay on A, not on its own.

F. Pairs trading — evidence: **weak**

Only conference papers and a preprint with short test periods; the one journal paper could not be read. It also needs shorting, which retail traders can do only intraday or through futures. Confidence: low.

G. Other ideas found

- **"Avoid" list from the weekly SELL side** (ALADIN's own evidence): the most reliable result in this whole audit; useful as an exit or no-buy filter, not as a source of profit on its own (shorting needs futures).
- **Low-volatility / low-risk**: contested in India (one study for, one against).
- **News and social sentiment**: Indian studies find it short-lived or not predictive; supports keeping sentiment out of the signal, as ALADIN's own tests also found.
- **Short-term reversal**: one weak study; costs are the big risk at daily and weekly holds.

What this means for Phase 4

Order of testing, strongest evidence first: **A (momentum, long-only) with E (risk control)**, then **C (machine learning at a 20-day hold)**, then quality and low-risk variants, then B, D and F as checks. Each will be tested at tight, medium and loose filters, as you asked, inside your −25% limit.